Analysis of Eurovision Song Contest

Statistical Learning Project

IBRAHIM KILIC

Agenda

1 Introduction

2 Exploratory Data Analysis

3 Data Analysis

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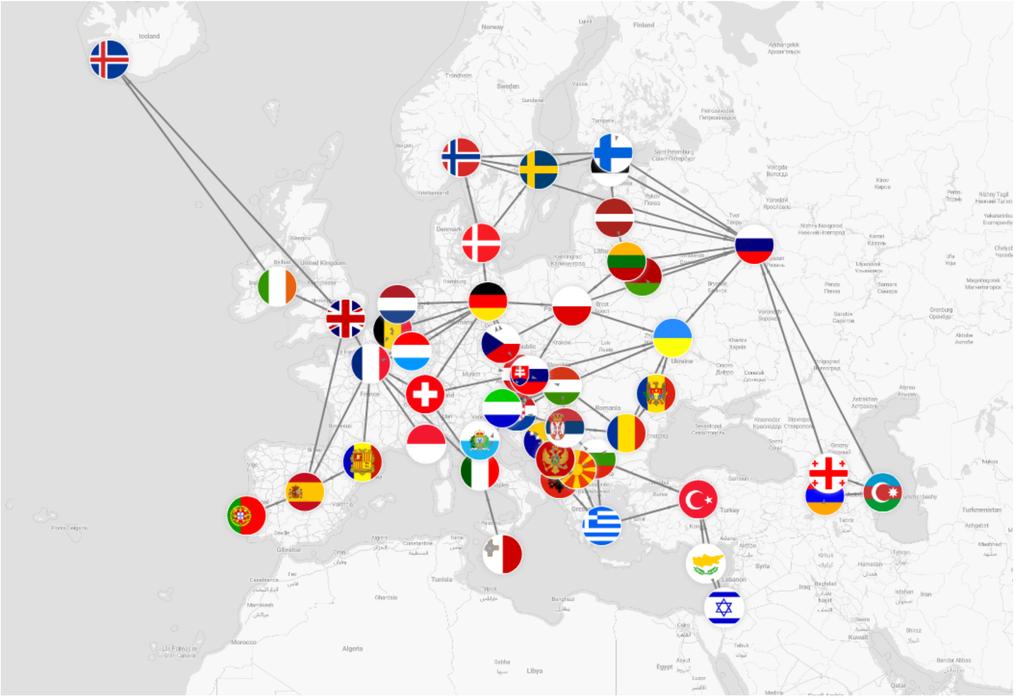
Introduction

- The Eurovision Song Contest is one of the most popular music events in Europe, attracting millions of viewers annually.
- This project aims to analyse the factors influencing the success of songs in the Eurovision Song Contest using statistical methods.
- The study will explore song characteristics, performance order, and voting outcomes to uncover insights into what contributes to a song's success in the competition.



Introduction

- **Stakeholders:** This project is relevant to a wide range of audiences interested in the Eurovision Song Contest: Eurovision Fans, Music Industry Professionals, Academics and General Public
- **Objective:** The primary goal of this project is to identify and analyse the factors that influence a song's success in the Eurovision Song Contest. Specifically, the project seeks to answer the following research questions:
    - How do factors such as song genre, tempo, language, and the artist's gender affect a song's ranking in the top 10?
    - What factors explain the differences between jury and public voting outcomes?
    - Does the running order have a significant impact on the final scores?

Introduction

Methodology: This project utilize the “song_data.csv" dataset and apply various statistical analysis methods, including: Logistic Regression, Decision Trees, Multiple Regression, Clustering Analysis, Regression Analysis

song_data

year	semi_final	semi_draw_position	final_draw_position	country	artist_name	song_name	language	style	direct_qualifier_10	gender_main_singers	age	selection	key	BPM	energy	danceability	happiness	loudness	acousticness	instrumentalness	liveness	speechiness	release_date	key_change_10	backing_dancers	backing_singers	backing_instruments	instrument_10	qualified_10	final_televote_points	final_jury_points	final_televote_votes	final_jury_votes	final_place	final_total_points	semi_place	semi_televote_points	semi_jury_points	semi_total_points	favourite_10	race	host_10
2025	1	1	20	Norway	Alessandra	Queen of Kings	English	Pop	0	Female	1	unknown	unknown	E Minor	110	36	64	23	10-dB	58	0	16	3	unknown	unknown	4	0	0	0	1	216	52	36	11	5	268	6	102	102	0	unknown	0
2025	1	2		Malta	The Busker	Dance (Our Own Party)	English	Pop	-	Male	1	unknown	unknown	F Minor	100	78	70	82	6-dB	2	0	18	4	unknown	unknown	0	0	2	0	0							15	3		3	0	unknown	0
2025	1	3	5	Serbia	Luke Black	Samo mi se spava	Serbian, English	Pop	0	Male	1	unknown	unknown	A Major	100	70	56	11	10-dB	4	2	32	5	unknown	unknown	4	0	0	0	1	16	14	4	6	24	30	10	37	37	0	unknown	0
2025	1	4		Latvia	Sudden Lights	Aja	English	Rock	-	Male	1	unknown	unknown	A Minor	160	55	56	40	6-dB	5	0	8	7	unknown	unknown	0	0	3	0	0							11	34		34	0	unknown	0
2025	1	5	2	Portugal	Minnical	Ai coração	Portuguese	Pop	0	Female	1	unknown	unknown	F♯ Minor	140	63	66	77	8-dB	31	0	16	5	unknown	unknown	4	0	0	0	1	16	43	3	9	23	59	9	74	74	0	unknown	0
2025	1	6		Ireland	Wild Youth	We Are One	English	Rock	-	Male	1	unknown	unknown	D Major	112	66	47	12	7-dB	4	0	9	3	unknown	unknown	0	0	3	0	0							12	10		10	0	unknown	0
2025	1	7	25	Croatia	Let 3	Mama ŠĆ!	Croatian	Pop	0	Male	1	unknown	unknown	C Major	130	78	60	63	7-dB	16	0	21	9	unknown	unknown	0	4	0	0	0	112	11	26	2	13	123	8	76	76	0	unknown	0
2025	1	8	3	Switzerland	Remo Forrer	Watergun	English	Ballad	0	Male	1	unknown	unknown	Ab Minor	130	87	68	39	5-dB	48	0	18	7	unknown	unknown	4	0	0	0	1	31	61	10	15	20	92	7	97	97	0	unknown	0
2025	1	9	23	Israel	Noa Kirel	Unicorn	English	Pop	0	Female	1	unknown	unknown	F♯ Minor	130	87	68	28	5-dB	2	0	32	10	unknown	unknown	5	0	0	0	1	185	177	38	25	3	362	3	127	127	0	unknown	0
2025	1	10	18	Moldova	Pasha Parfeni	Soarele și luna	Romanian	Traditional	0	Male	1	unknown	unknown	Ab Major	120	86	62	34	5-dB	36	0	18	14	unknown	unknown	0	2	3	0	1	76	20	17	4	18	96	5	109	109	0	unknown	0
2025	1	11	9	Sweden	Loreen	Tattoo	English	Pop	0	Female	1	unknown	unknown	E♭ Minor	150	78	55	30	6-dB	24	0	12	8	unknown	unknown	0	0	0	0	1	243	340	36	36	1	583	2	135	135	1	unknown	0
2025	1	12		Azerbaijan	TuraTuranX	Tell Me More	English	Pop	-	Male	2	unknown	unknown	D Major	96	75	71	42	8-dB	2	0	29	4	unknown	unknown	0	0	0	1	0							14	4		4	0	unknown	0
2025	1	13	14	Czech Republic	Vesna	My Sister's Crown	English, Ukrainian, Czech, Bulgarian	Pop	0	Female	6	unknown	unknown	E Minor	100	92	55	51	6-dB	5	0	37	16	unknown	unknown	0	0	0	0	1	35	94	12	18	10	129	4	110	110	0	unknown	0
2025	1	14		Netherlands	Mia Nicole and Dion Cooper	Burning Daylight	English	Ballad	-	Mix	2	unknown	unknown	E♭ Major	71	43	28	29	9-dB	58	0	11	4	unknown	unknown	0	0	0	0	0							13	7		7	0	unknown	0
2025	1	15	13	Finland	Kaliroji	Cha Cha Cha	Finnish	Pop	0	Male	1	unknown	unknown	D Major	150	79	72	74	6-dB	0	0	11	4	unknown	unknown	4	0	0	0	1	376	150	37	22	2	526	1	177	177	0	unknown	0
2025	2	1		Denmark	Reiley	Breaking My Heart	English	Pop	-	Male	1	unknown	unknown	D Major	194	75	59	57	6-dB	3	0	10	36	unknown	unknown	0	4	0	0	0							14	4		4	0	unknown	0
2025	2	2	17	Armenia	Brunette	Future Lover	English, Armenian	Ballad	0	Female	1	unknown	unknown	D Minor	100	44	56	24	11 dB	89	0	10	4	unknown	unknown	0	0	0	0	1	53	69	10	15	14	122	6	99	99	0	unknown	0
2025	2	3		Romania	Theodor Andrei	O.G.T. (Off and On)	Romanian, English	Pop	-	Male	1	unknown	unknown	A Minor	110	75	68	84	6-dB	22	0	25	3	unknown	unknown	0	0	0	1	0							15	0		0	0	unknown	0
2025	2	4	12	Estonia	Akka	Bridges	English	Ballad	0	Female	1	unknown	unknown	C Minor	140	47	59	33	5-dB	84	0	14	3	unknown	unknown	0	0	0	1	0	22	146	4	22	8	168	10	74	74	0	unknown	0
2025	2	5	16	Belgium	Gustaph	Because of You	English	Pop	0	Male	1	unknown	unknown	C Minor	127	91	74	7	6-dB	0	82	16	4	unknown	unknown	1	3	0	0	1	55	127	13	22	7	182	8	90	90	0	unknown	0
2025	2	6	7	Cyprus	Andrew Lambrou	Break a Broken Heart	English	Ballad	0	Male	1	unknown	unknown	E♭ Minor	130	74	43	43	4-dB	6	0	19	4	unknown	unknown	0	0	0	0	1	58	68	11	17	12	126	7	84	84	0	unknown	0
2025	2	7		Iceland	Diljá	Power	English	Pop	-	Female	1	unknown	unknown	F Minor	160	73	41	27	5-dB	1	0	35	5	unknown	unknown	0	0	0	0	0							11	44		44	0	unknown	0
2025	2	8		Greece	Victor Vernicos	What They Say	English	Pop	-	Male	1	unknown	unknown	C Major	154	55	40	27	6-dB	31	0	28	11	unknown	unknown	0	0	0	0	0							13	14		14	0	unknown	0
2025	2	9	4	Poland	Blanka	Solo	English	Pop	0	Female	1	unknown	unknown	B Minor	99	64	90	72	7-dB	18	0	12	4	unknown	unknown	4	0	0	0	1	81	12	17	5	19	93	3	124	124	0	unknown	0
2025	2	10	24	Slovenia	Joker Out	Carpe Diem	Slovene	Rock	0	Male	1	unknown	unknown	Cs Minor	132	66	88	56	4-dB	1	0	16	4	unknown	unknown	0	0	4	0	1	45	33	11	6	21	78	5	103	103	0	unknown	0
2025	2	11		Georgia	Iru	Echo	English	Ballad	-	Female	1	unknown	unknown	G Minor	140	91	59	37	4-dB	11	2	36	14	unknown	unknown	0	0	0	0	0							16	33		33	0	unknown	0
2025	2	12		San Marino	Piqued Jacks	Like an Animal	English	Rock	-	Male	1	unknown	unknown	C Major	128	79	54	64	6-dB	0	0	21	4	unknown	unknown	0	0	3	0	0							16	0		0	0	unknown	0
2025	2	13	1	Austria	Teya and Salena	Who the Hell Is Edgar?	English	Pop	0	Female	2	unknown	unknown	B Minor	146	64	69	51	8-dB	2	0	10	6	unknown	unknown	4	0	0	0	1	16	164	4	18	15	120	2	137	137	0	unknown	0
2025	2	14	10	Albania	Albina and Familja Kelmendi	Duje	Albanian	Traditional	0	Female	1	unknown	unknown	D Minor	110	80	64	43	6-dB	5	0	13	19	unknown	unknown	0	5	0	0	1	59	17	16	4	22	76	9	83	83	0	unknown	0
2025	2	15	22	Lithuania	Monika Linkytė	Stay	English	Ballad	0	Female	1	unknown	unknown	B Major	98	55	40	18	7-dB	3	0	8	4	unknown	unknown	0	4	0	0	1	46	81	8	15	11	127	4	110	110	0	unknown	0
2025	2	16	15	Australia	Voyager	Promise	English	Rock	0	Male	1	unknown	unknown	C Major	122	70	62	33	6-dB	0	0	10	7	unknown	unknown	0	0	4	0	1	21	130	6	23	9	151	1	149	149	0	unknown	0
2025	-	-	6	France	La Zarra	Évidemment	French	Ballad	1	Female	1	unknown	unknown	B Major	120	75	73	23	5-dB	12	0	8	7	unknown	unknown	0	0	0	0	0	50	54	17	10	16	104				0	unknown	0
2025	-	-	8	Spain	Blanca Paloma	Eaea	Spanish	Traditional	1	Female	1	unknown	unknown	F Minor	116	73	61	26	4-dB	7	0	39	4	unknown	unknown	5	0	0	0	0	5	95	2	21	17	190				0	unknown	0
2025	-	-	11	Italy	Marco Mengoni	Due vite	Italian	Ballad	1	Male	1	unknown	unknown	D Major	117	54	64	13	6-dB	40	0	9	3	unknown	unknown	2	0	0	0	0	174	176	29	27	4	350				0	unknown	0
2025	-	-	19	Ukraine	Tvorchi	Heart of Steel	English, Ukrainian	Pop	1	Male	1	unknown	unknown	Ab Minor	76	52	53	34	8-dB	1	0	9	5	unknown	unknown	2	0	0	1	0	189	54	28	10	8	243				0	unknown	0
2025	-	-	21	Germany	Lord of the Lost	Blood & Glitter	English	Rock	1	Male	1	unknown	unknown	A Minor	139	80	64	47	4-dB	1	0	5	8	unknown	unknown	0	0	4	1	-	15	3	3	2	26	18				0	unknown	0
2025	-	-	26	United Kingdom	Mae Muller	I Wrote a Song	English	Pop	1	Female	1	unknown	unknown	A Minor	125	89	71	86	4-dB	4	0	8	5	unknown	unknown	0	4	0	0	-	9	15	2	5	25	24				0	unknown	1
2025	1	1		Albania	Ronela Hajati	Sekret	Albanian, English	Pop	-	Female	1	unknown	unknown	F Minor	100	70	76	45	6-dB	5	0	32	5	unknown	unknown	5	0	0	0	0							12	46	12	58	0	unknown	0
2025	1	2		Latvia	Citi Zeni	Eat Your Salad	English	Pop	-	Male	1	unknown	unknown	B Minor	115	69	75	59	6-dB	1	0	34	17	unknown	unknown	0	0	4	0	0							14	16	39	55	0	unknown	0
2025	1	3	14	Lithuania	Monika Liu	Sentimental	Lithuanian	Ballad	0	Female	1	unknown	unknown	C Major	93	56	68	50	8-dB	9	0	31	3	unknown	unknown	0	0	0	0	1	93	35	21	10	14	128	7	103	56	159	0	unknown	0
2025	1	4	5	Switzerland	Marius Bear	Boys Do Cry	English	Ballad	0	Male	1	unknown	unknown	F Major	116	18	42	39	11 dB	87	0	11	4	unknown	unknown	0	0	0	0	1	0	78	6	21	17	78	8	11	107	118	0	unknown	0
2025	1	5	-	Slovenia	LPS	Disko	Slovene	Pop	-	Male	1	unknown	unknown	E Minor	120	61	68	60	6-dB	0	0	5	3	unknown	unknown	0	0	4	0	0							17	8	7	15	0	unknown	0
2025	1	6	12	Ukraine	Kalush Orchestra	Stefania	Ukrainian	Traditional	0	Male	1	unknown	unknown	D Major	100	82	83	32	5-dB	14	0	12	4	unknown	unknown	1	2	2	0	1	439	192	39	26	1	631	1	202	135	337	1	unknown	0
2025	1	7	-	Bulgaria	Intelligent Music Project	Intention	English	Rock	-	Male	1	unknown	unknown	D Major	80	84	52	72	4-dB	3	2	24	3	unknown	unknown	0	0	0	0	0							16	18	11	29	0	unknown	0
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2022	1	10	3	Portugal	Maro	Saudade, saudade	English, Portuguese	Ballad	0	Female	6	unknown	unknown	Bb Minor	83	34	64	38	11 dB	68	0	11	4	unknown	unknown	0	0	0	0	1	36	171	9	29	9	207	4	87	171	258	0	unknown	0
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2022	1	15	17	Greece	Amanda Georgiadi Tenfjord	Die Together	English	Ballad	0	Female	1	unknown	unknown	F Major	150	37	48	20	8-dB	12	0	13	7	unknown	unknown	0	0	0	0	1	57	158	9	26	8	215	3	60	151	211	0	unknown	0
2022	1	16	7	Norway	Subwoolfer	Give That Wolf a Banana	English	Pop	0	Male	2	unknown	unknown	G Minor	124	53	74	55	10-dB	20	0	8	4	unknown	unknown	4	0	0	0	1	146	36	32	10	10	182	6	104	73	177	0	unknown	0
2022	1	17	8	Armenia	Rosa Linn	Snap	English	Ballad	0	Female	1	unknown	unknown	C Major	170	64	56	53	6-dB	11	0	45	6	unknown	unknown	0	0	1	1	1	21	40	4	10	20	81	5	105	82	187	0	unknown	0
2022	2	1	4	Finland	The Rasmus	Jezebel	English	Rock	0	Male	1	unknown	unknown	B Minor	116	87	50	49	4-dB	0	0	8	4	unknown	unknown	0	0	3	0	1	26	12	8	3	21	38	7	99	63	162	0	unknown	0
2022	2	2	-	Israel	Michael Ben David	I.M	English	Pop	-	Male	1	unknown	unknown	Ab Minor	120	82	70	46	7-dB	1	0	13	6	unknown	unknown	4	0	0	0	0							13	27	34	61	0	unknown	0
2022	2	3	24	Serbia	Konstrakta	In corpore sano	Serbian, Latin	Traditional	0	Female	1	unknown	unknown	C Major	136	71	82	63	7-dB	1	0	17	8	unknown	unknown	5	0	0	0	1	225	87	34	15	5	312	3	174	63	237	0	unknown	0
2022	2	4	15	Azerbaijan	Nadir Rustamli	Fade to Black	English	Ballad	0	Male	1	unknown	unknown	A Minor	79	30	34	30	10-dB	19	0	14	6	unknown	unknown	1	0	0	0	1	3	103	1	19	16	106	10	0	96	96	0	unknown	0
2022	2	5	-	Georgia	Circus Mircus	Lock Me In	English	Rock	-	Male	1	unknown	unknown	D Major	107	76	79	84	9-dB	0	0	36	4	unknown	unknown	0	0	0	1	0							18	9	13	22	0	unknown	0
2022	2	6	-	Malta	Emma Muscat	I Am What I Am	English	Pop	-	Female	1	unknown	unknown	G Major	96	71	56	38	5-dB	4	0	9	3	unknown	unknown	4	0	0	1	0							16	20	27	47	0	unknown	0
2022	2	7	-	San Marino	Achille Lauro	Stripper	Italian, English	Rock	-	Male	1	unknown	unknown	A Major	140	80	60	55	6-dB	0	0	8	10	unknown	unknown	0	0	5	0	0							14	29	21	50	0	unknown	0
2022	2	8	21	Australia	Sheldon Riley	Not the Same	English	Ballad	0	Male	1	unknown	unknown	Bb Minor	128	43	36	38	10-dB	66	0	14	4	unknown	unknown	0	0	0	0	1	2	123	1	23	15	125	2	74	169	243	0	unknown	0
2022	2	9	-	Cyprus	Andromache	Ela	English, Greek	Pop	-	Female	1	unknown	unknown	Eb Minor	96	63	66	65	7-dB	24	0	6	9	unknown	unknown	2	0	0	0	1							12	54	9	63	0	unknown	0
2022	2	10	-	Ireland	Brooke	That's Rich	English	Pop	-	Female	1	unknown	unknown	A Major	122	60	68	56	4-dB	0	0	34	3	unknown	unknown	4	0	0	0	0							15	35	12	47	0	unknown	0
2022	2	11	-	North Macedonia	Andrea	Circles	English	Ballad	-	Female	1	unknown	unknown	Cs Minor	144	71	54	22	5-dB	48	0	17	3	unknown	unknown	0	0	0	0	0							11	20	56	76	0	unknown	0

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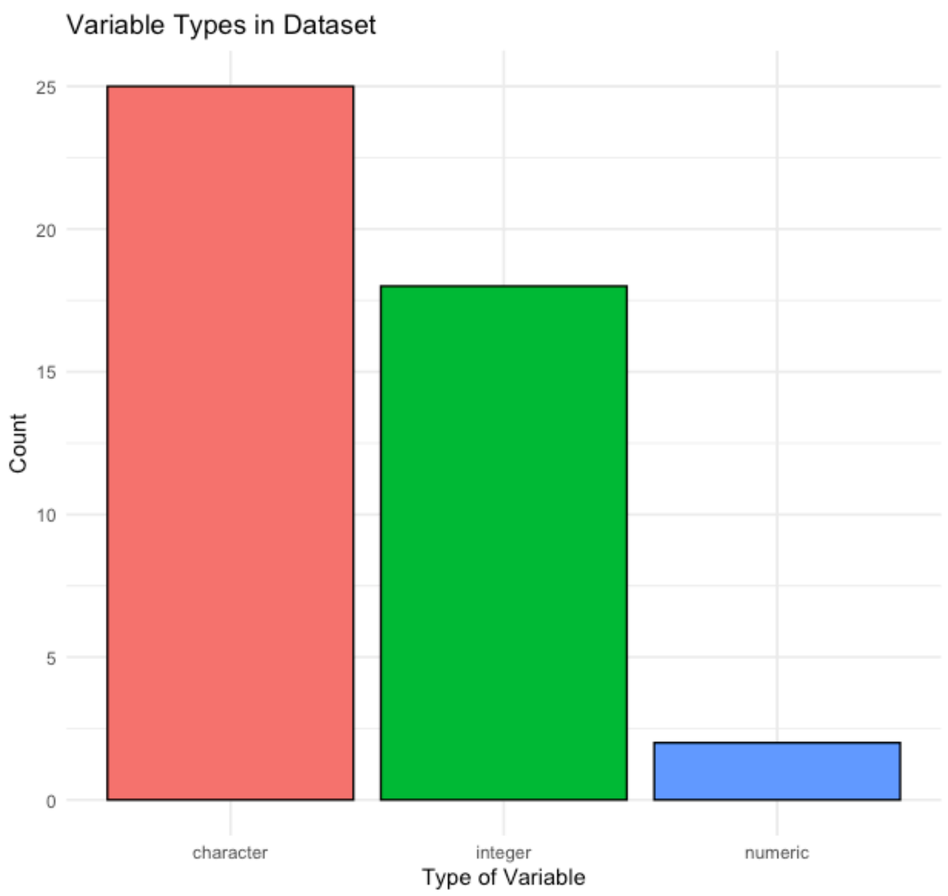
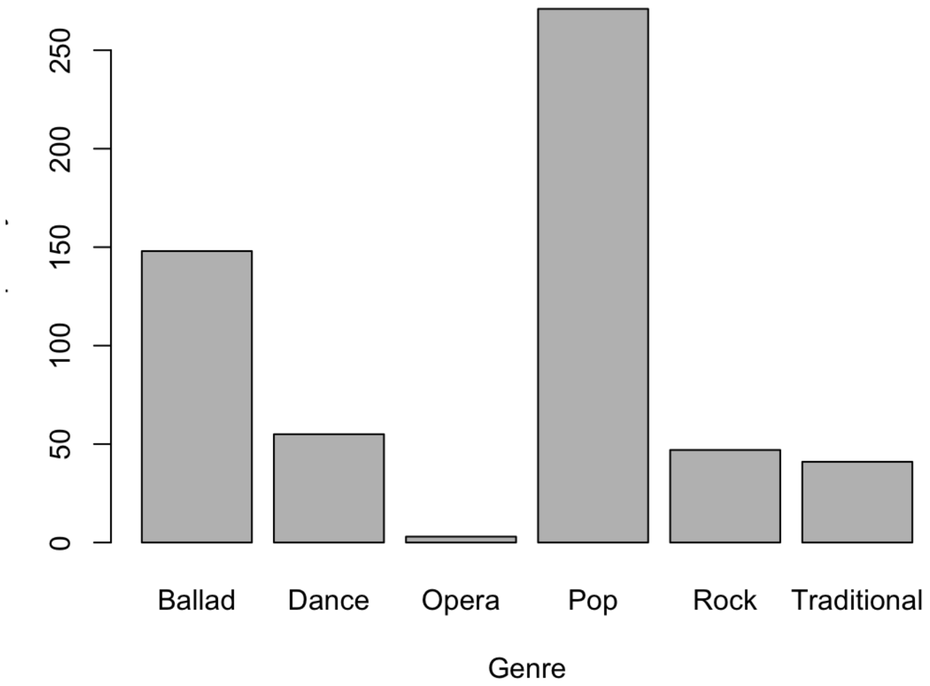
Exploratory Data Analysis

Source: The raw data employed in this project was downloaded from the Kaggle.

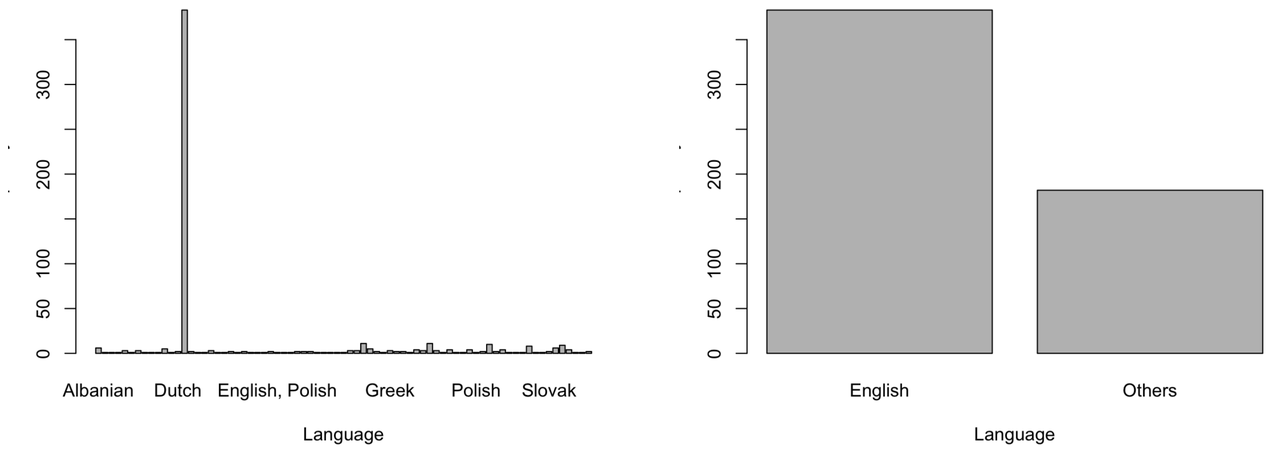
Structure: 565 observations, 44 variables (both numerical and categorical)



Exploratory Data Analysis



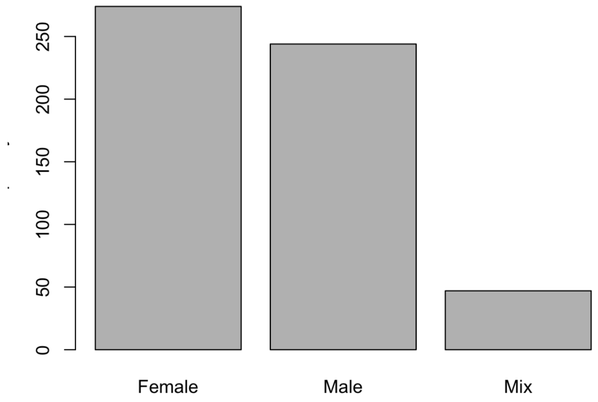
Genre distribution



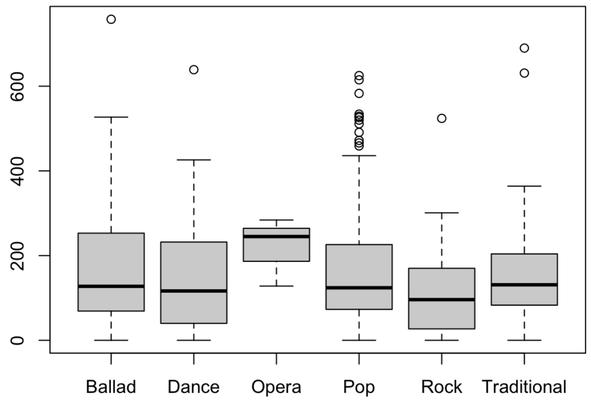
Language distribution

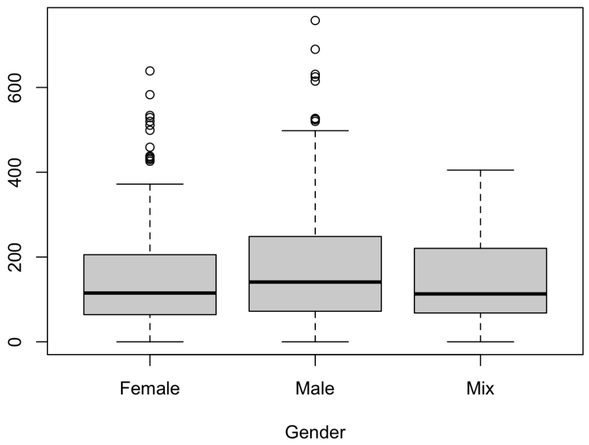
Exploratory Data Analysis

Gender distribution

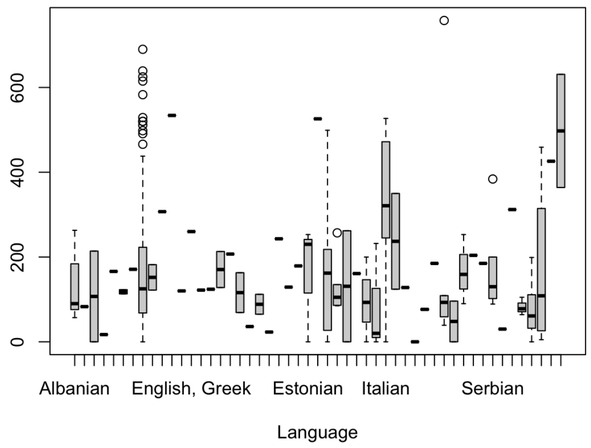


Total Points by genre



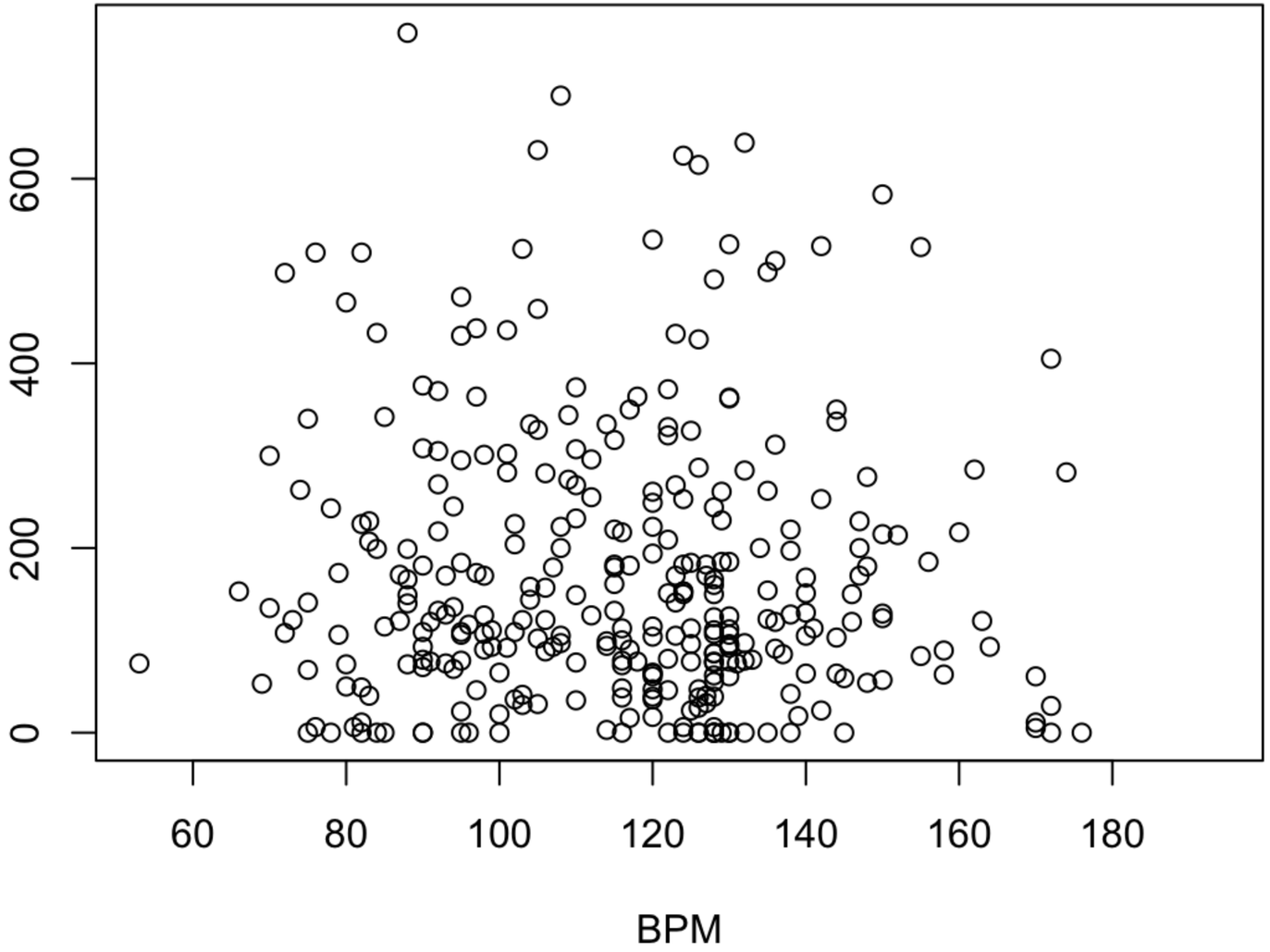


Total Points by gender



Total Points by language

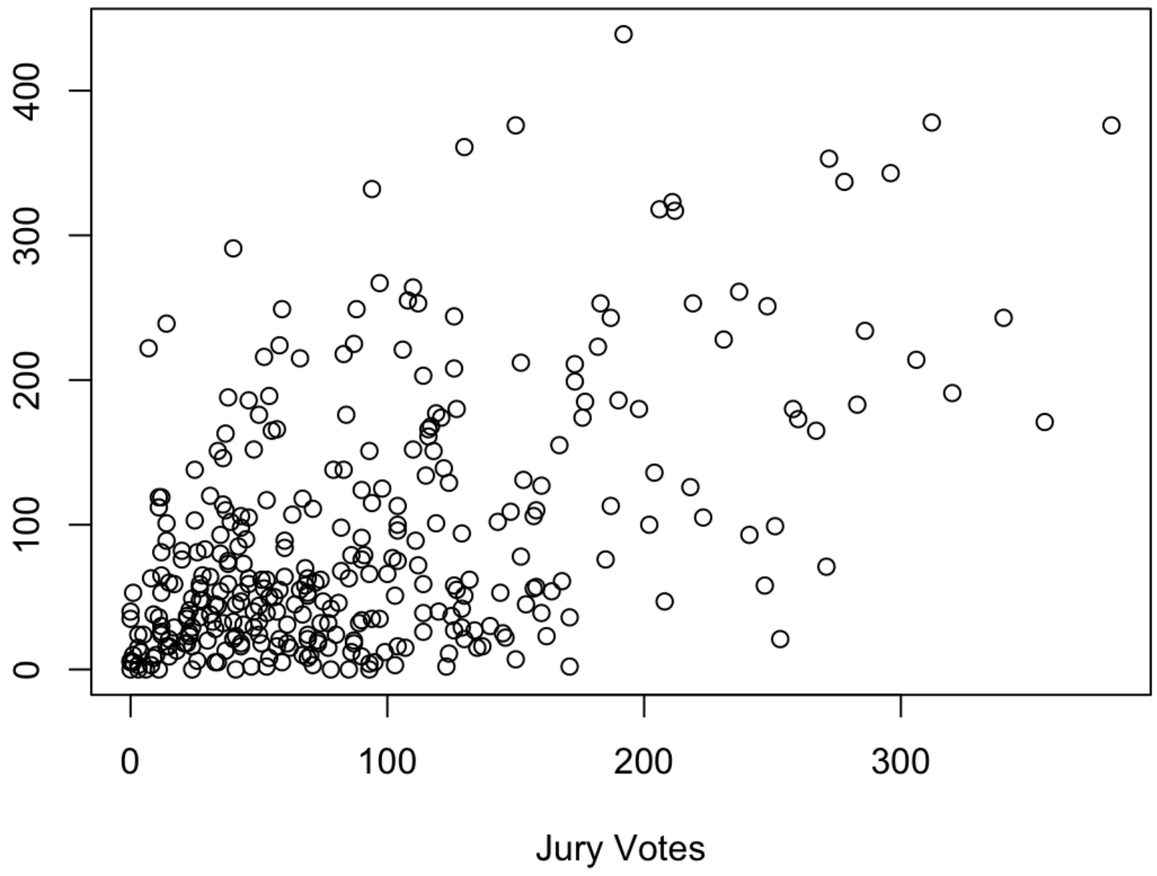
Exploratory Data Analysis



-0.05137087

Correlation between BPM and success score

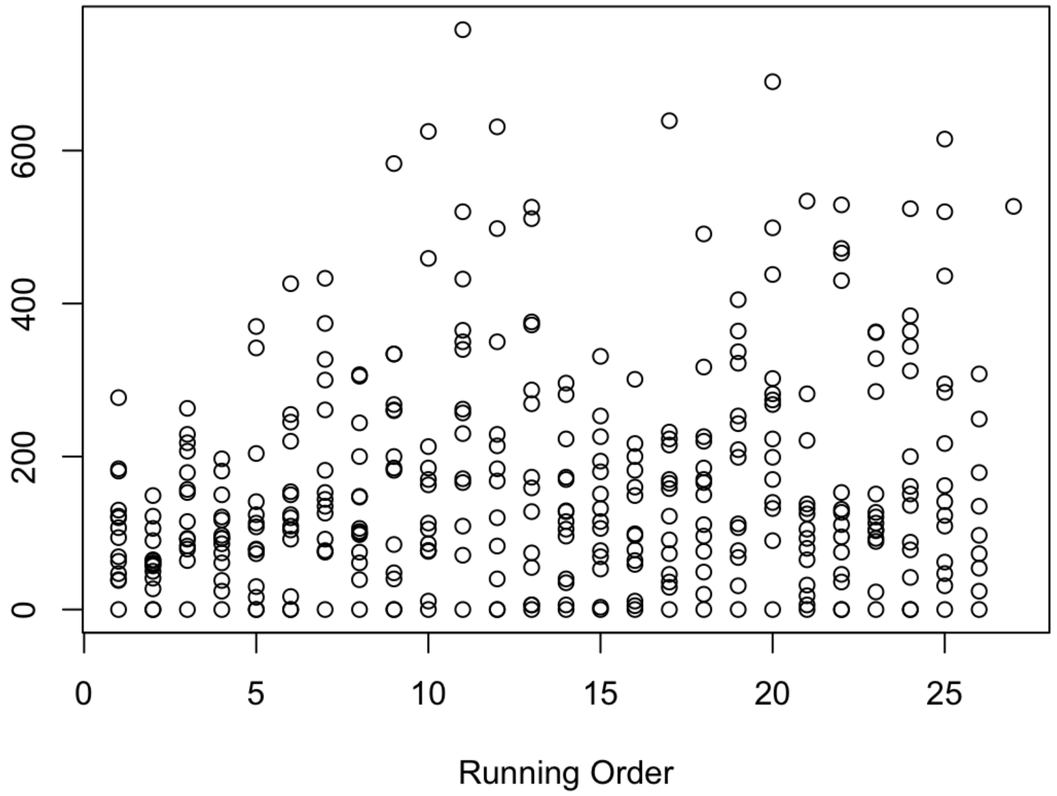
Exploratory Data Analysis



0.5239011

Correlation between jury votes and public votes

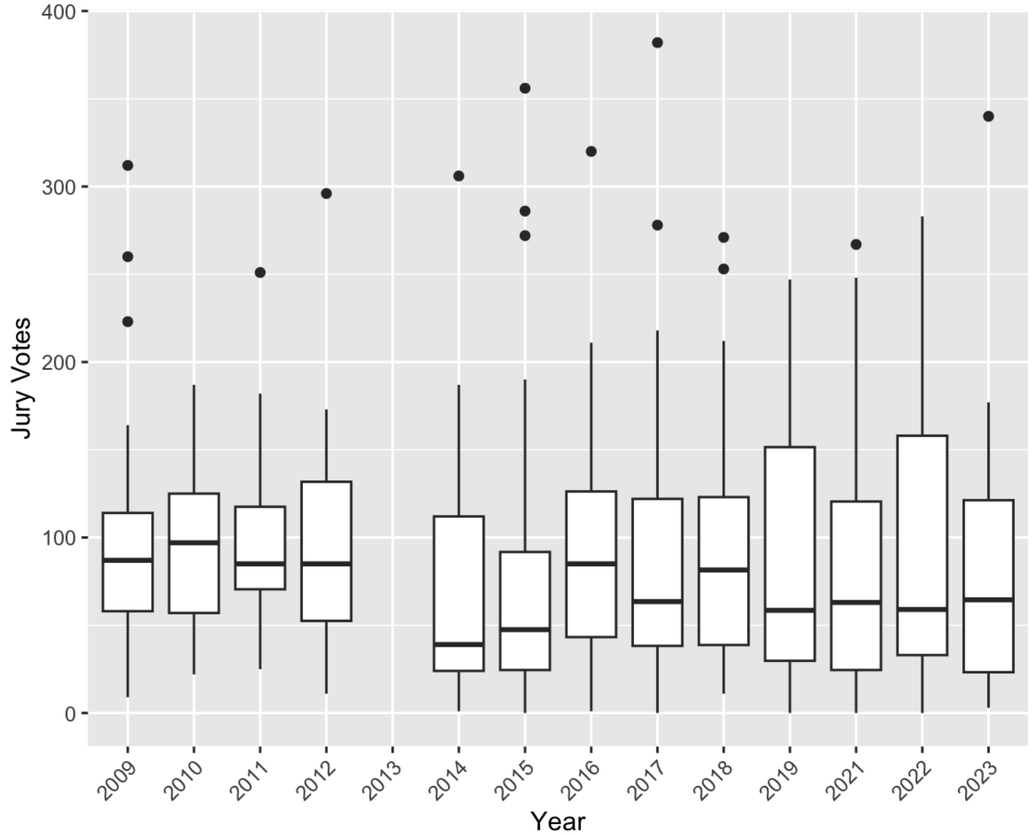
Exploratory Data Analysis

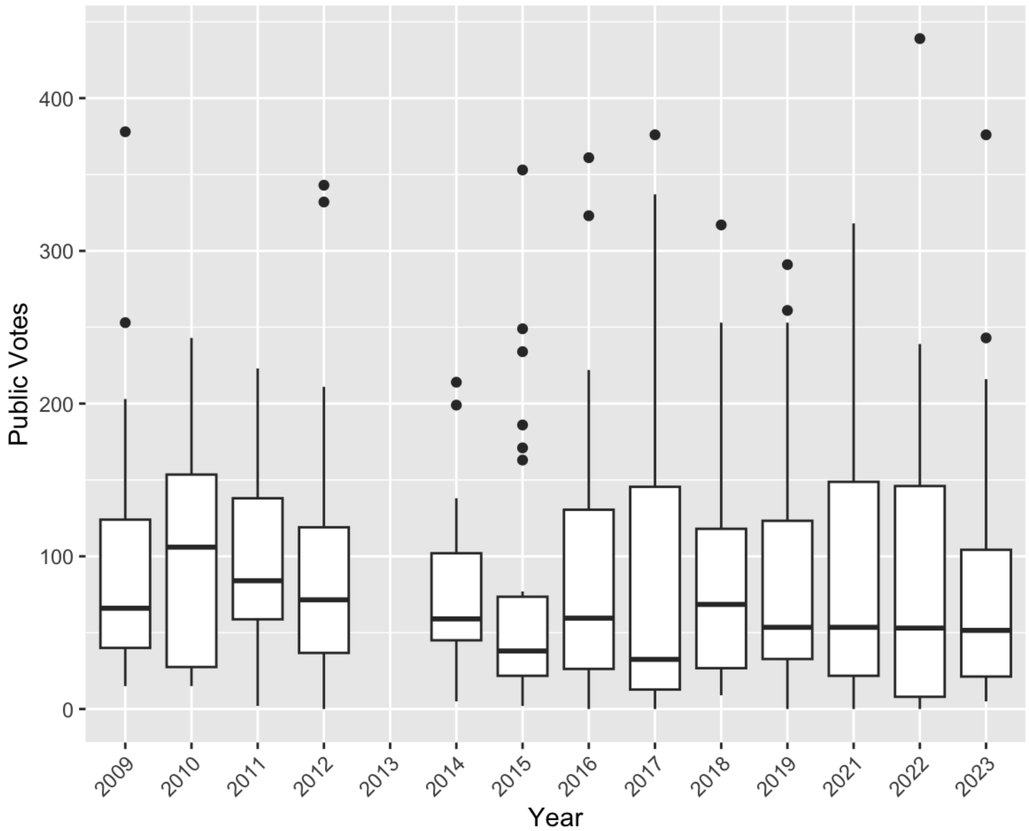


Relationship between running order and total points

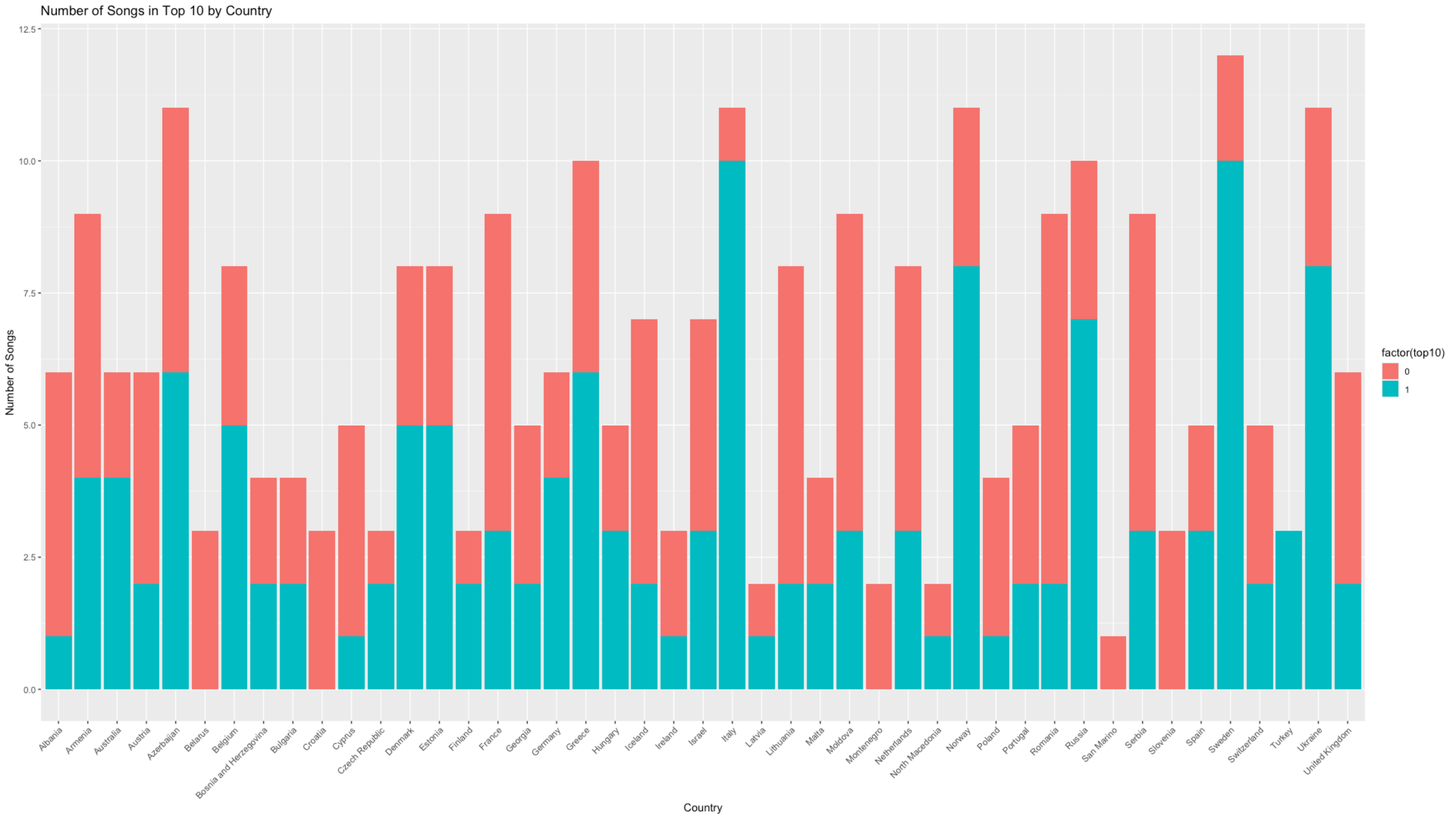
Exploratory Data Analysis

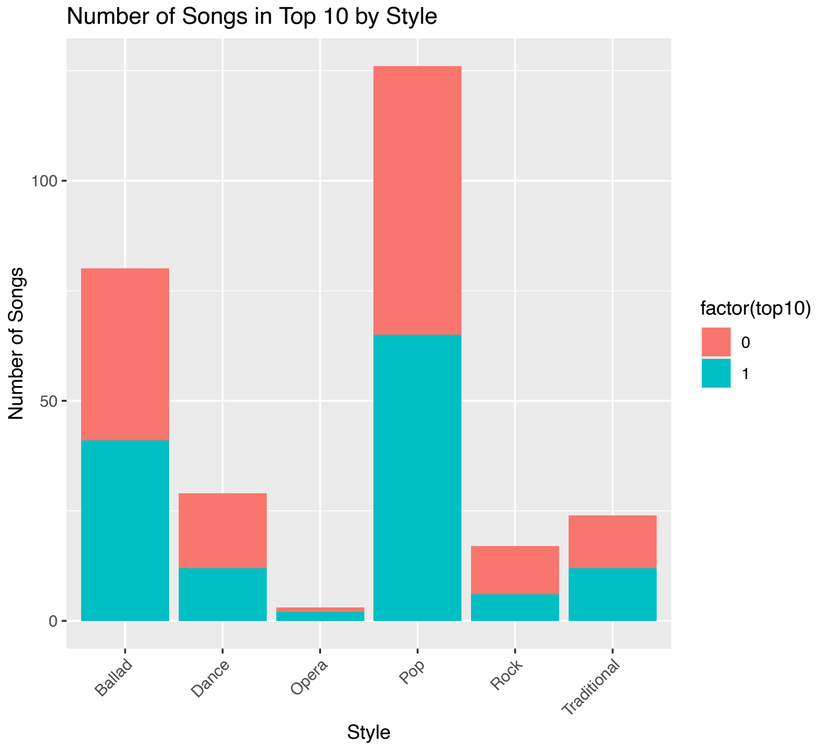
Time Series Analysis

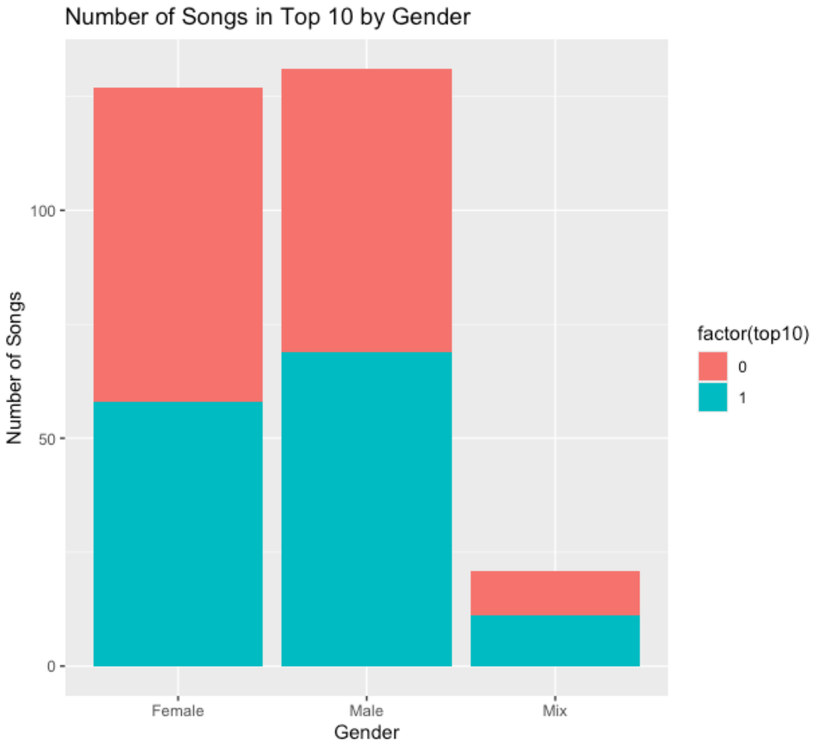


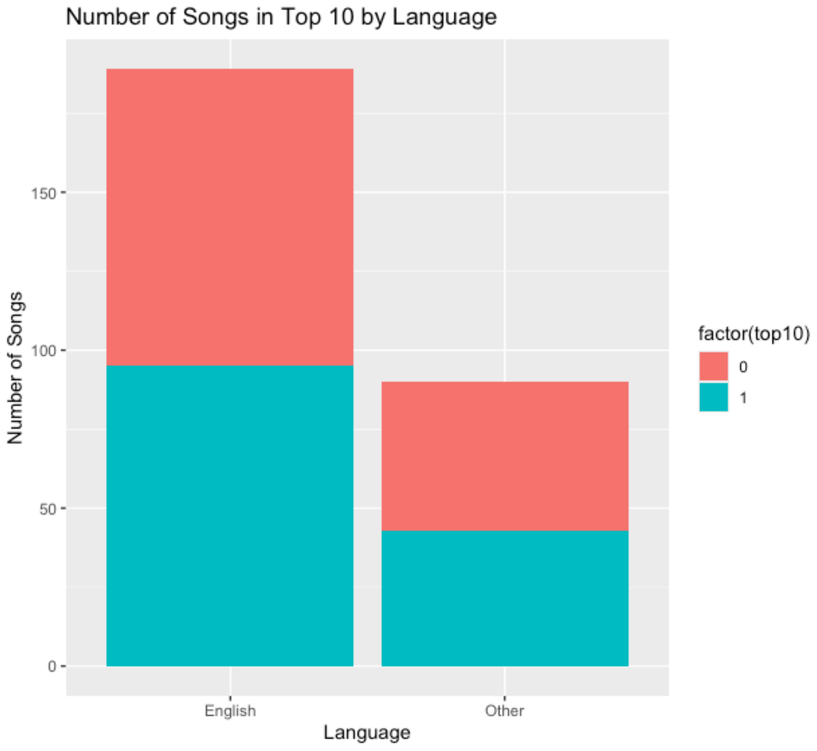


Change in jury and public votes over the years


Number of Songs in Top 10 by Country
Number of Songs
12.5
10.0
7.5
5.0
2.5
0.0
Albania
Armenia
Australia
Austria
Azerbaijan
Belarus
Belgium
Bosnia and Herzegovina
Bulgaria
Croatia
Cyprus
Czech Republic
Denmark
Estonia
Finland
France
Georgia
Germany
Greece
Hungary
Iceland
Ireland
Israel
Italy
Latvia
Lithuania
Malta
Moldova
Montenegro
Netherlands
North Macedonia
Norway
Poland
Portugal
Romania
Russia
San Marino
Serbia
Slovenia
Spain
Sweden
Switzerland
Turkey
Ukraine
United Kingdom
Country
factor(top10)
0
1


Number of Songs in Top 10 by Style
Number of Songs
100
50
0
Ballad
Dance
Opera
Pop
Rock
Traditional
Style
factor(top10)
0
1


Number of Songs in Top 10 by Gender
Number of Songs
100
50
0
Female
Male
Mix
Gender
factor(top10)
0
1


Number of Songs in Top 10 by Language
Number of Songs
150
100
50
0
English
Other
Language
factor(top10)
0
1

Agenda

1 Introduction

2 Exploratory Data Analysis

3 Data Analysis

4 Conclusion

Data Analysis

- **Analysis 1:** How do factors such as song genre, BPM, language, and the artist's gender affect a song's ranking in the top 10?
- The first is a binary classification problem where the goal is to estimate how likely a song in the dataset would stand a chance to be in top 10 at the final of Eurovision Song Contest. For this reason, **Logistic Regression** and **Decision Tree models** is used.

Data Analysis Analysis 1: **Logistic Regression**

Model Fitting:

- The logistic regression model is performed in R environment.
- The dependent variable is defined as successful which is represented by; Success while Independent variables include; s*tyle, BPM, language*, and *gender*.

Data Analysis Analysis 1: **Logistic Regression**

Prediction and Evaluation:

On the test dataset, the trained model will make some predictions then determine how well the model has performed or done.

Genre	In Top 10	Out of Top 10
Ballad	39	41
Dance	17	12
Opera	1	2
Pop	61	65
Rock	11	6
Traditional	12	12

Residual Deviance	AIC	Logistic Regression Accuracy
269.1083	447.1083	0.738351254480287

Data Analysis Analysis 1: **Logistic Regression**

Prediction and Evaluation:

.

Out of Top 10

	Female	Male	Mix
Ballad	20	15	4
Dance	9	7	1
Opera	0	1	0
Pop	32	25	4
Rock	2	9	0
Traditional	6	5	1

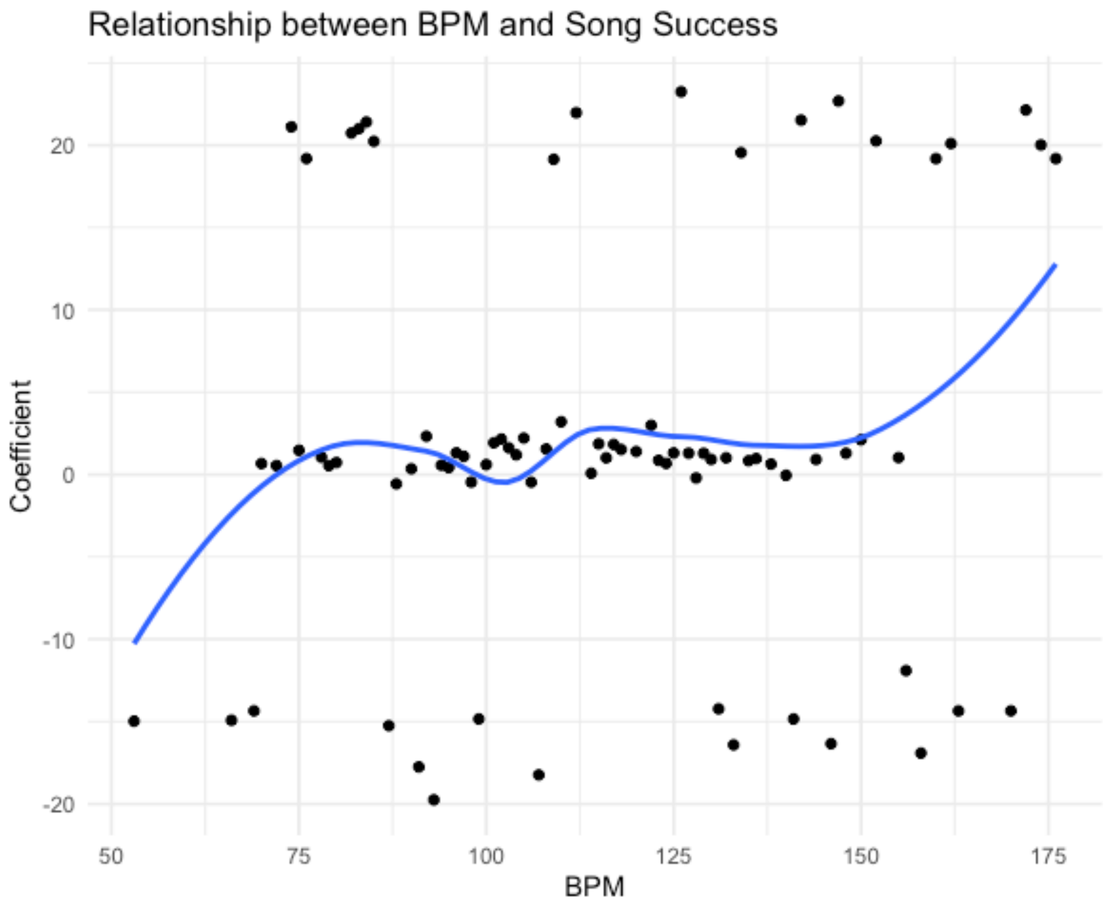
In Top 10

	Female	Male	Mix
Ballad	21	18	2
Dance	5	5	2
Opera	2	0	0
Pop	25	34	6
Rock	1	5	0
Traditional	4	7	1

Data Analysis Analysis 1: **Logistic Regression**

Prediction and Evaluation:

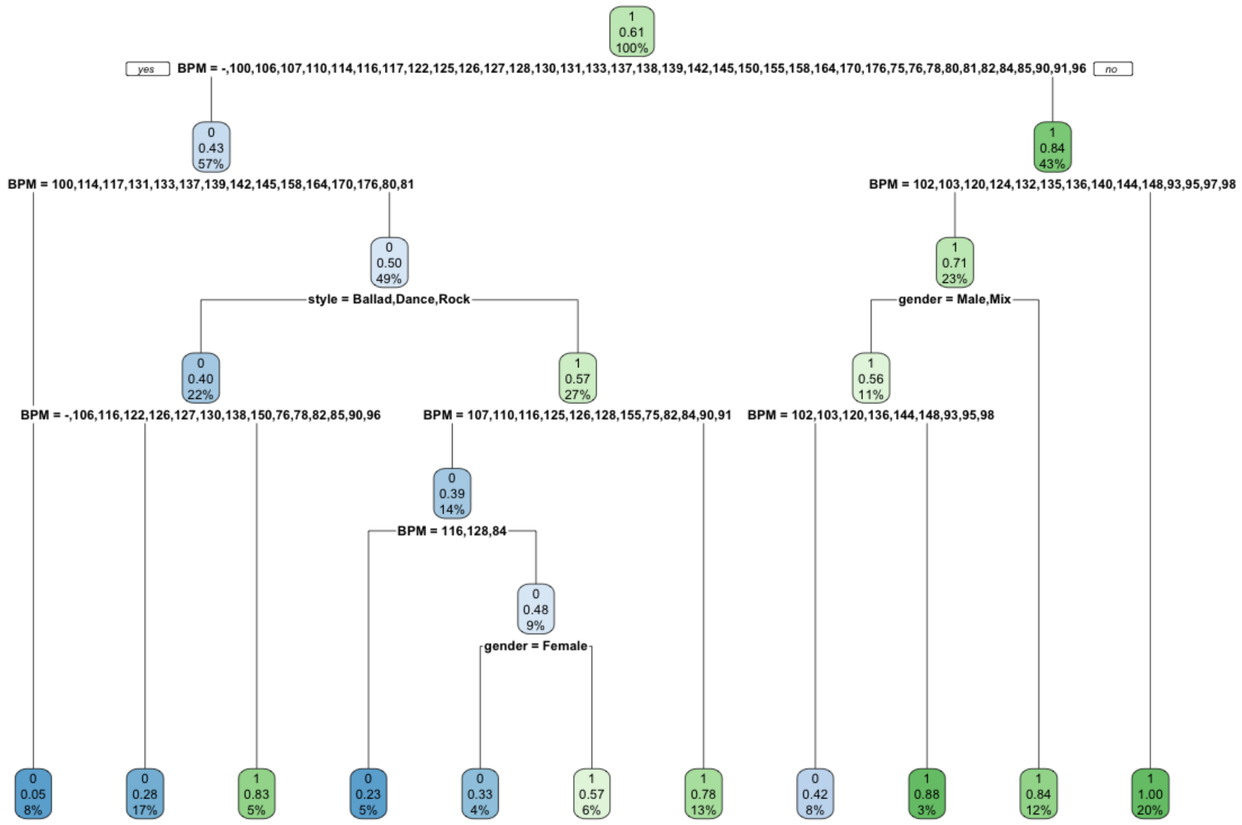
.



Data Analysis Analysis 1: **Decision Tree**

Model Fitting:

- The Decision Tree model is fitted As with any logistic regression analysis, the dependent variable is Success while the independent variables include *style, BPM, language*, and *gender*.



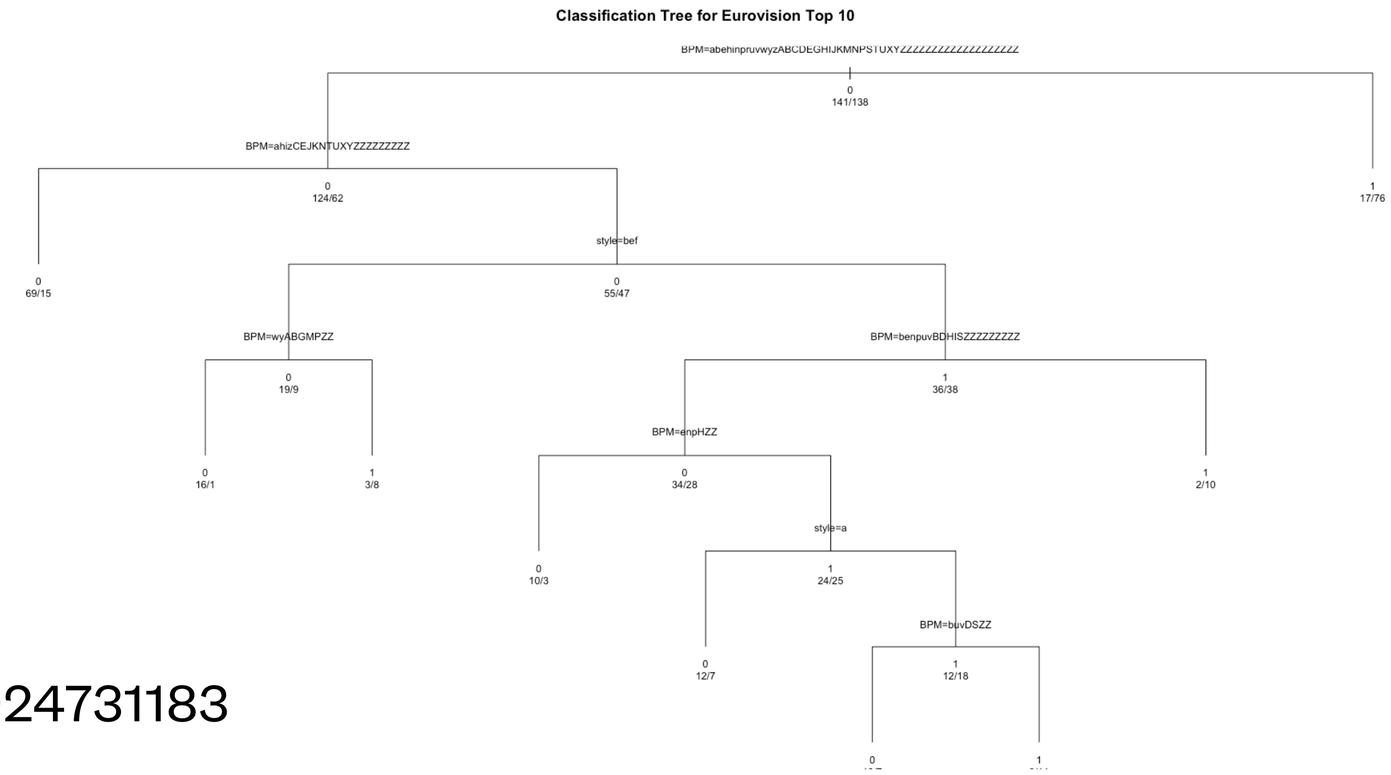
Data Analysis Analysis 1: **Decision Tree**

Prediction and Evaluation:

Variable performance

BPM	Style	Gender
93	5	2

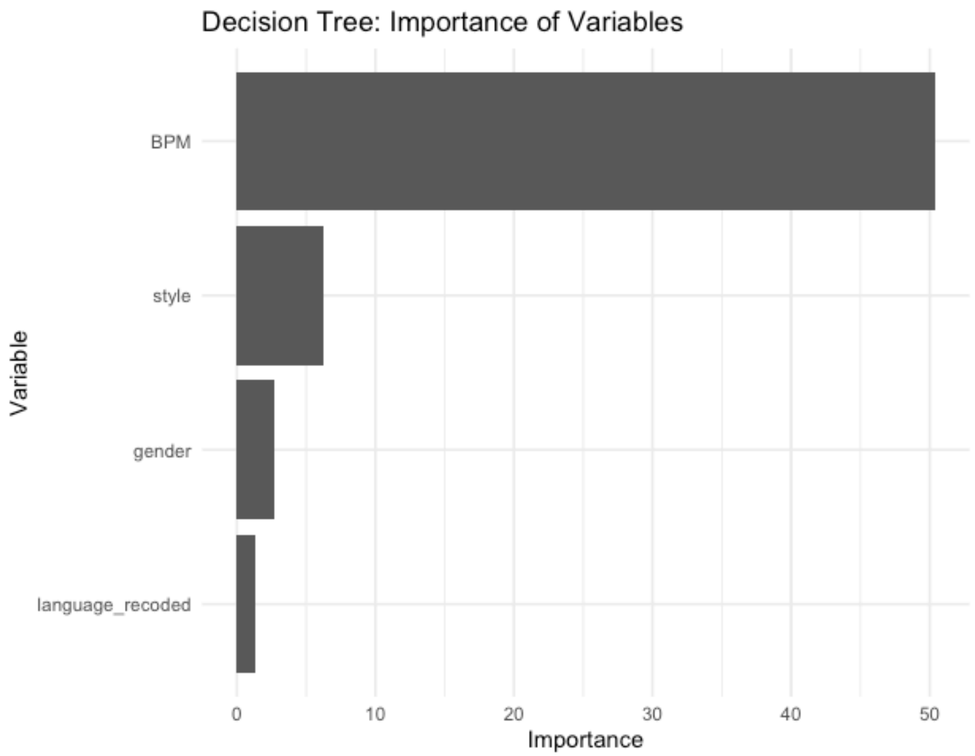
Decision Tree Accuracy: 0.795698924731183



Data Analysis Analysis 1: **Decision Tree**

Prediction and Evaluation:

.

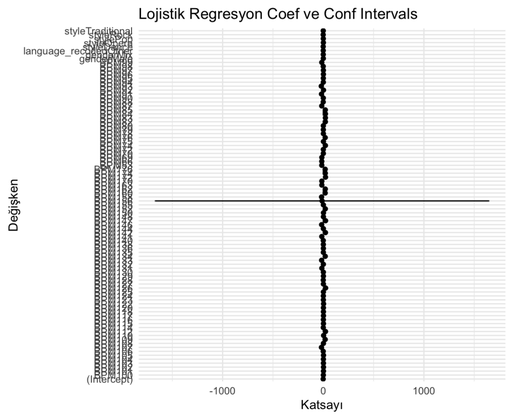


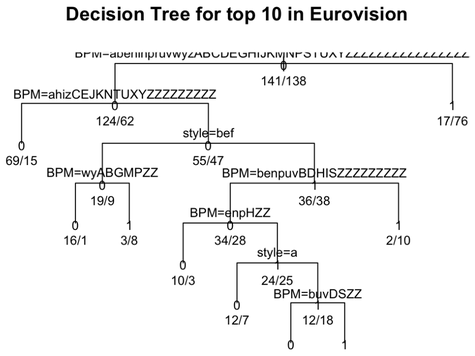
Data Analysis

Analysis 1: **Logistic Regression** vs. **Decision Tree**

Logistic Regression

Sensitivity	Specificity	Accuracy
0.710144927536232	0.765957446808511	0.738351254480287



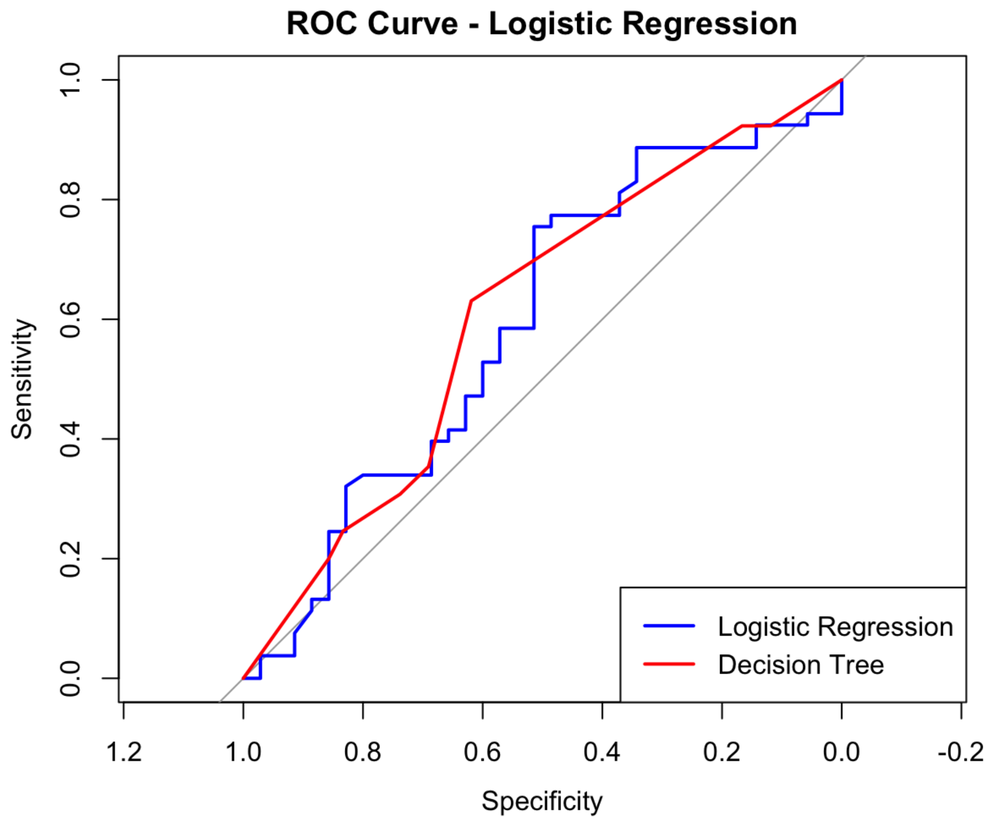


Decision Tree

Sensitivity	Specificity	Accuracy
0.760869565217391	0.829787234042553	0.795698924731183

Data Analysis

Analysis 1: **Logistic Regression** vs. **Decision Tree**



Data Analysis Analysis 2

- In this analysis, the differences between jury and public voting in the Eurovision Song Contest were analysed. factors influencing these votes were identified, and countries with similar voting patterns were clustered.
- **Methods:**
  - Multiple Regression: To determine the factors influencing jury and public votes.
  - Principal Component Analysis (PCA): To reduce data dimensionality and visualise voting patterns.
  - Clustering Analysis: To group countries based on similarities in their voting behavior.

Data Analysis Analysis 2: **Multiple Regression**

- Fit separate multiple regression models for jury and public votes.
- Dependent variables: Jury votes and public votes.
- Independent variables: style, country, gender and language
- Analyse the coefficients and p-values to identify significant predictors.

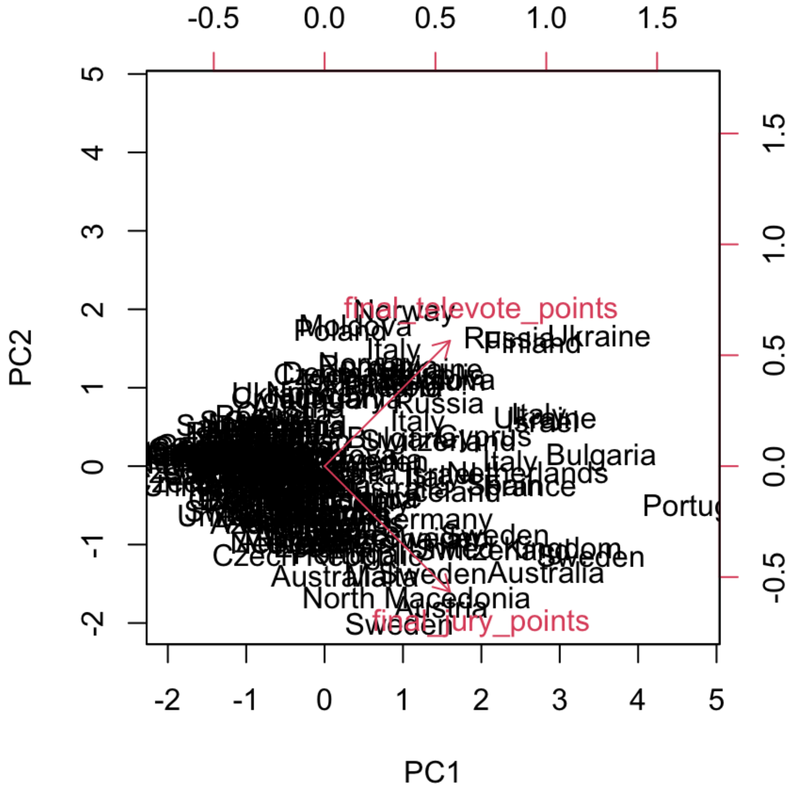
Data Analysis Analysis 2: **Principal Component Analysis (PCA)**

- PCA applied to the jury and televote data to reduce dimensionality and explore the variance in jury and televote points data."
- Principal Component 1 (PC1) explains 74.79% of the variance in the data.
- Principal Component 2 (PC2) explains 25.21% of the variance in the data.
- Together, **PC1 and PC2** account for **100%** of the variance.

	PC1	PC2
Standard Deviation	1.2230	0.7101
Proportion of Variance	0.7479	0.2521
Cumulative Proportion	0.7479	1.0000

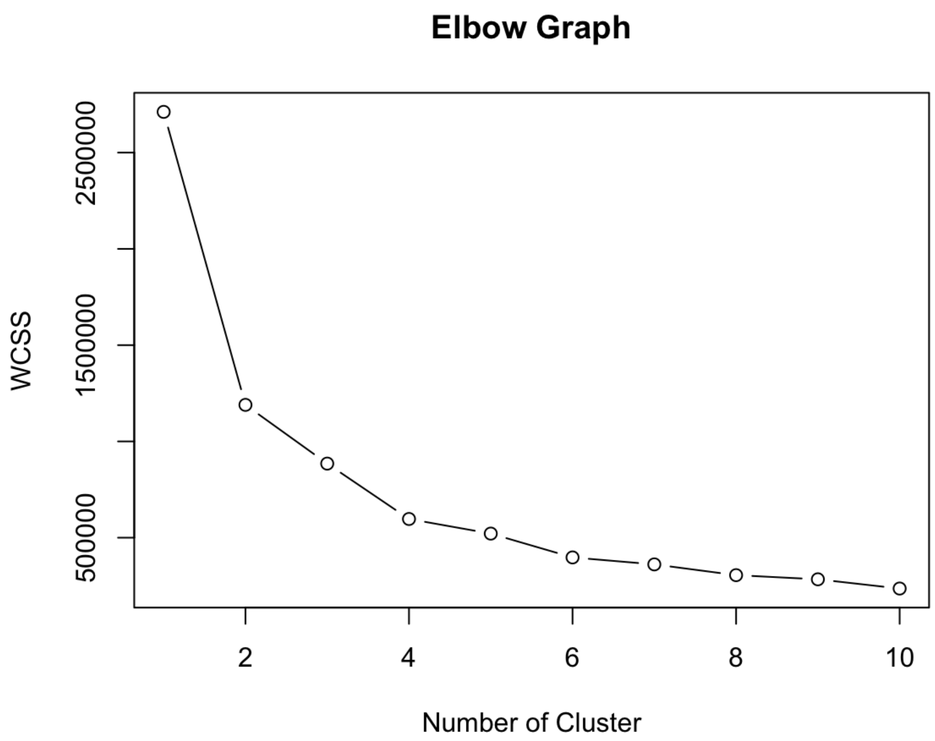
Data Analysis

Analysis 2: **Principal Component Analysis (PCA)**



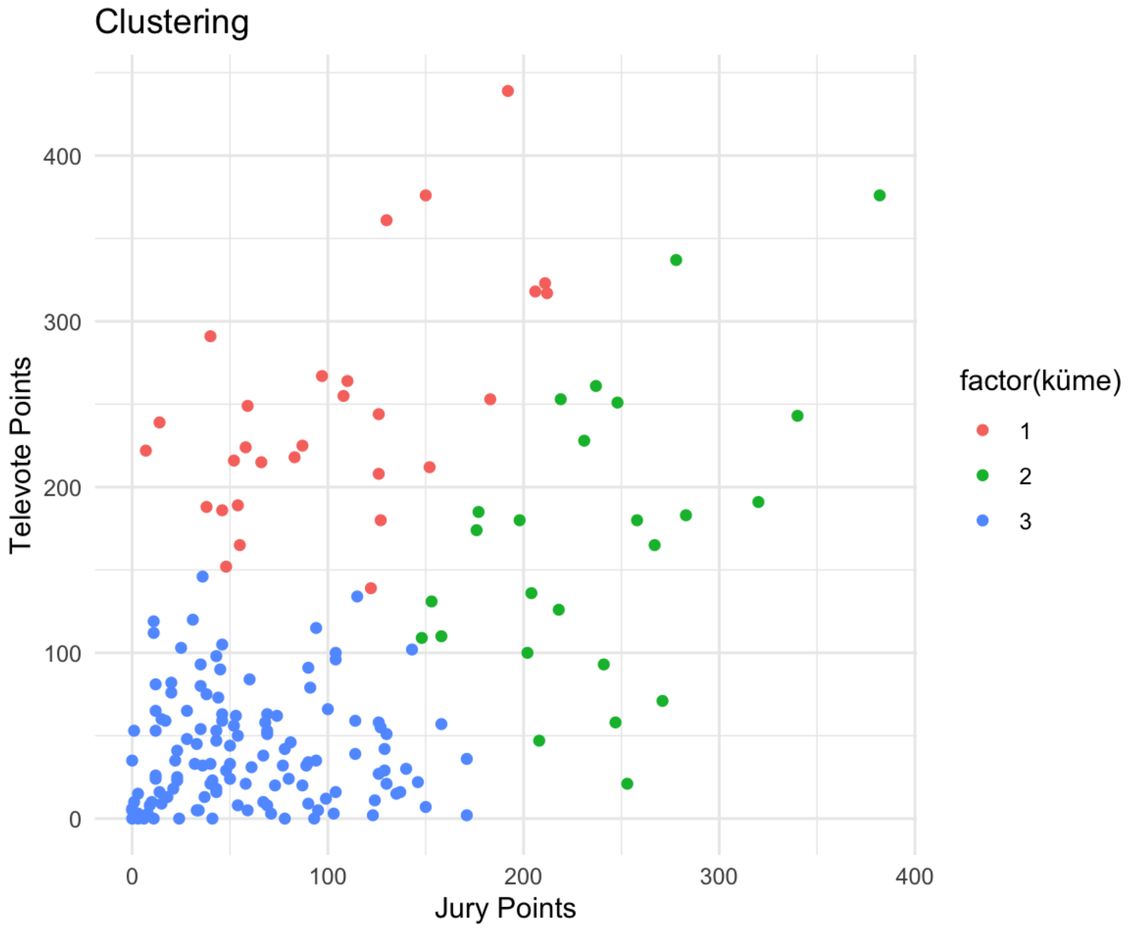
Data Analysis Analysis 2: **K Means Clustering**

- K-Means clustering was applied with the optimal number of clusters.
- From elbow graph, number of cluster is chosen as 3.
- The results were added to the dataset as a new column.



Data Analysis Analysis 2: **K Means Clustering**

- K-Means clustering results



Data Analysis Analysis 3

- To explore the relationship between final draw position and points received in Eurovision final, focusing on televote and jury scores.
- Model Fitting:
  - Used **Linear Regression** to assess the relationship between: Final draw position and televote points, Final draw position and jury points.
  - **Hypotheses Testing** is used to evaluate regression coefficients for significance using t-tests. Conducted an ANOVA to validate model significance.

Data Analysis Analysis 3: **Hypotheses Testing**

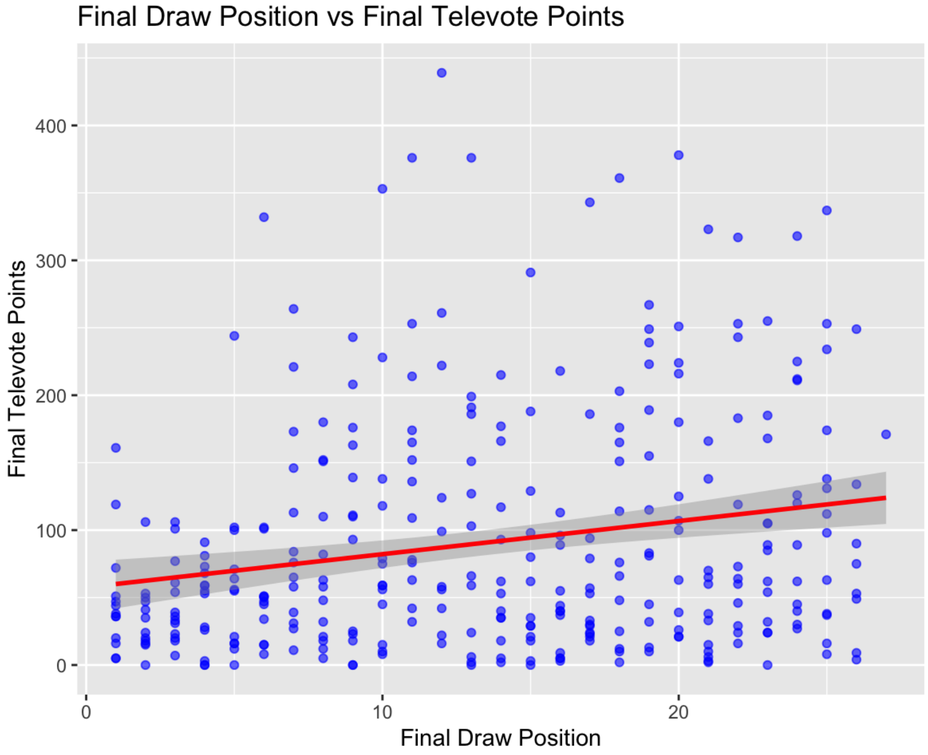
Final Televote Points

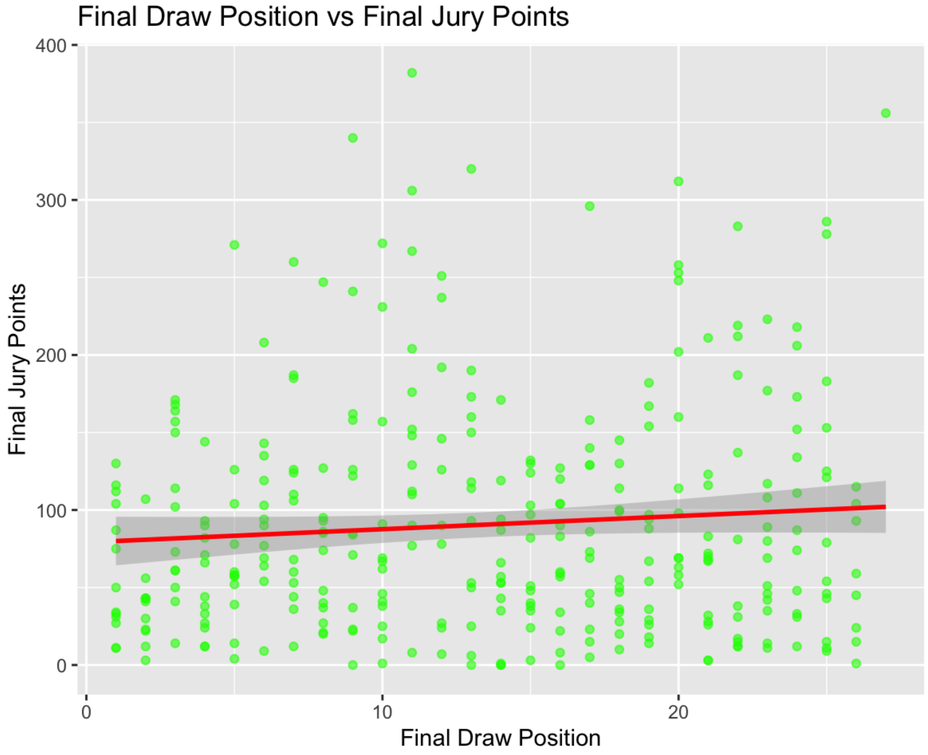
	Df	Sum Sq	Mean Sq	F value
Final Draw Position	1	108520	108520	14.999
Residuals	324	2344118	7235	

Final Jury Points

	Df	Sum Sq	Mean Sq	F value
Final Draw Position	1	12918	12918.1	2.3674
Residuals	324	1767997	5456.8	

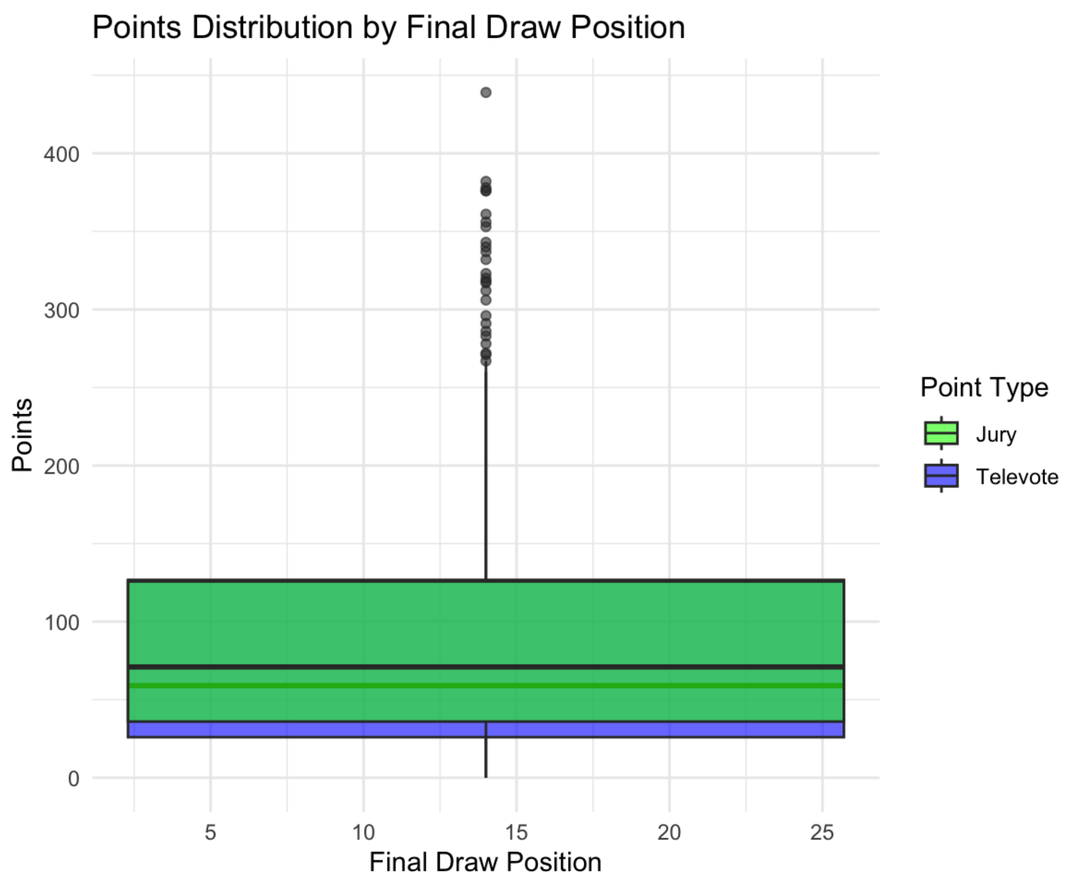
Data Analysis Analysis 3: **Linear Regression**





	Model	Slope	p_value
	Televote	2.4581447	0.0001301173
	Jury	0.8481109	0.1248727573

Data Analysis Analysis 3: **Linear Regression**



Data Analysis Analysis 3

Prediction and Evaluation:

- Televote Points:
    - Draw position had a statistically significant effect ($p<0.001$)
    - The model explained ~4.4% of the variance in televote points ($R^2 = 0.044$)
- Jury Points:
    - No statistically significant relationship ($p=0.125$)
    - The model explained only 0.7% of the variance in jury points ($R^2 = 0.007$)
- Key Observations:
    - Maximum televote points were awarded to a performance in position 12.
    - Maximum jury points were awarded to a performance in position 11.

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Conclusion

- Based on the first analysis, decision tree is more flexible, captures the pattern better although both methods (logistic regression model) have provide interpretability.
- Decision tree is better in terms of sensitivity, specificity, and accuracy.
- Logistic regression offers simpler framework but it’s linearity may limit it’s performance.

Conclusion

- PCA reveals that most of the variance in the voting data is explained by a single component (PC1), suggesting a dominant factor influencing the voting patterns.
- K-Means clustering uncovers three distinct patterns in the voting data, which could represent different voting behaviors or styles among countries.
- Final draw position has a small but significant impact on televote points, while its influence on jury points is negligible.
- Future research could incorporate additional predictors like performance quality, genre, or stage effects to better understand scoring patterns.